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Operating System

6Week [14 October - 20 October]

SOC3010 OS QUIZ3 FALL 2024

Started on	2024-10-18 12:54
State	Finished
Completed on	2024-10-26 23:59
Time taken	8 days 11 hours
Marks	665.00/910.00
Grade	7.31 out of 10.00 (73%)

Question 1

Incorrect

Mark 0.00 out of 10.00

In Muti-APIC system, each Core has two Interrupt pins. What are they

Select one:

- ☐ 1. LINT1, LINT2
- ☐ 2. LINT0, LINT1
- ☐ 3. INT1, INT2
- ☒ 4. INTO, INT1 ✖

The correct answer is: LINT0, LINT1

Question 2

Incorrect

Mark 0.00 out of 5.00

..... is a Processor Detected Exception

Select one:

- ☐ 1. int 0x55
- ☒ 2. int 0x80 ✖
- ☐ 3. int 0x48
- ☐ 4. Fault

The correct answer is: Fault

Question 3

Correct

Mark 5.00 out of 5.00

The function is invoked to execute all interrupt service routines associated with an interrupt.

Answer:



The correct answer is: do_IRQ()

Course Info ▲

Syllabus

Participants list

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Question 4

Correct

Mark 5.00 out of 5.00

The int vector associated with the exception "Divide Error" is

Select one:

- ☒ 1. 0 ✓
- ☐ 2. 14
- ☐ 3. 8
- ☐ 4. 4

The correct answer is: 0

Question 5

Correct

Mark 5.00 out of 5.00

Keyboard Controller Status Register port address is

Select one:

- ☐ 1. 0x60
- ☐ 2. 0x84
- ☐ 3. 0x80
- ☒ 4. 0x64 ✓

The correct answer is: 0x64

Question 6

Incorrect

Mark 0.00 out of 5.00

In Linux , overflow exception handler is activated using which of the following :

Select one:

- ☐ 1. System gate
- ☐ 2. System interrupt gate
- ☐ 3. Interrupt gate
- ☒ 4. Trap gate ✗

The correct answer is: System gate

Question 7

Correct

Mark 5.00 out of 5.00

Kernel makes use of Hard IRQ stack for each CPU in the system and each stack is contained in a single page frame.

Select one:

- ☐ 1. 3
- ☐ 2. 4
- ☐ 3. 2
- ☒ 4. 1 ✓

The correct answer is: 1

Question **8**
Correct
Mark 10.00 out of 10.00

..... Loglevel in printk indicates
"Normal message to take note of"

Select one:

- ☐ 1. KERN_WARNING
- ☐ 2. KERN_INFO
- ☒ 3. KERN_NOTICE ✓
- ☐ 4. KERN_DEBUG

The correct answer is:
KERN_NOTICE

Question **9**
Correct
Mark 5.00 out of 5.00

Assume that in Linux system, the processor raised "Double Fault" Exception when executing an instruction.
Which entry of the IDT is accessed?

Answer: ✓

The correct answer is: 8

Question **10**
Correct
Mark 5.00 out of 5.00

..... is a Programmed exception

Select one:

- ☐ 1. Fault
- ☐ 2. Trap
- ☒ 3. int 0x80 ✓
- ☐ 4. Abort

The correct answer is: int 0x80

Question **11**
Correct
Mark 5.00 out of 5.00

.....Post-interrupt processing of SCSI commands

Select one:

- ☐ 1. Timer_SOFTIRQ
- ☐ 2. NET_TX_SOFTIRQ
- ☒ 3. SCSI_SOFTIRQ ✓
- ☐ 4. TASKLET_SOFTIRQ

The correct answer is: SCSI_SOFTIRQ

Question **12**

Correct

Mark 5.00 out of 5.00

Kernel makes use of, exception stack for each process in the system

Select one:

- ☐ 1. 2
- ☐ 2. 4
- ☐ 3. 3
- ☒ 4. 1 ✓

The correct answer is: 1

Question **13**

Correct

Mark 5.00 out of 5.00

..... field in irqaction descriptor points to the interrupt service routine for an I/O device. This is the key field that allows many devices to share the same IRQ.

Answer: ✓

The correct answer is: handler

Question **14**

Correct

Mark 5.00 out of 5.00

..... actions are executed by the interrupt handler immediately, with the interrupts enabled.

Answer: ✓

The correct answer is: NonCritical

Question **15**

Correct

Mark 5.00 out of 5.00

..... is used to receive packets from network cards

Select one:

- ☐ 1. SCSI_SOFTIRQ
- ☒ 2. NET_RX_SOFTIRQ ✓
- ☐ 3. NET_TX_SOFTIRQ
- ☐ 4. TASKLET_SOFTIRQ

The correct answer is: NET_RX_SOFTIRQ

Question **16**
Correct
Mark 10.00 out of 10.00

..... of the same type are always serialized: in other words, the same type of cannot be executed by two CPUs at the same time.

Write only one answer

Answer: Tasklets



The correct answer is: Tasklets

Question **17**
Correct
Mark 10.00 out of 10.00

The IRQ affinity of a given interrupt also can be changed by the system administrator by writing a new CPU bitmap mask into the file (where n being the interrupt vector) .

Answer: /proc/irq/n/smp_affinity



The correct answer is: /proc/irq/n/smp_affinity

Question **18**
Correct
Mark 5.00 out of 5.00

If the interrupt is generated by the execution of an interrupt instruction in the program then it is

Select one:

- ☒ 1. synchronous
- ☐ 2. asynchronous
- ☐ 3. unexpected
- ☐ 4. unhandleable

The correct answer is: synchronous

Question **19**
Incorrect
Mark 0.00 out of 5.00

If two or more CPUs share the lowest priority, the interrupt requests are distributed between them using a technique called

Answer: round-robin



The correct answer is: arbitration

Question **20**

Correct

Mark 5.00 out of 5.00

The int vector associated with the exception "Double fault" is

Select one:

- ☒ 1. 8 ✓
- ☐ 2. 4
- ☐ 3. 5
- ☐ 4. 14

The correct answer is: 8

Question **21**

Correct

Mark 5.00 out of 5.00

..... Handles regular tasklets

Select one:

- ☐ 1. Timer_SOFTIRQ
- ☒ 2. TASKLET_SOFTIRQ ✓
- ☐ 3. NET_TX_SOFTIRQ
- ☐ 4. HI_SOFTIRQ

The correct answer is: TASKLET_SOFTIRQ

Question **22**

Incorrect

Mark 0.00 out of 5.00

..... actions are performed by means of separate functions called as "Softirqs and Tasklets"

Answer: Softirqs and Tasklets



The correct answer is: Noncritical deferrable

Question **23**

Correct

Mark 5.00 out of 5.00

Double fault exception is handled usinggate descriptor

Select one:

- ☒ 1. Task ✓
- ☐ 2. Trap
- ☐ 3. System Interrupt
- ☐ 4. Interrupt

The correct answer is: Task

Question **24**
Correct
Mark 10.00 out of 10.00

When the module is removed with `rmmod`, macro will be invoked, which will call the `__exit` function.

Answer: ✓

The correct answer is: `module_exit`

Question **25**
Correct
Mark 5.00 out of 5.00

During System bootstrap, all CPUs execute the function, which takes care of initializing the local APICs.

Select one:

- ☐ 1. `do_irq_balance()`
- ☐ 2. `do_IRQ()`
- ☐ 3. `set_ioapic_affinity_irq()`
- ☒ 4. `setup_local_APIC()` ✓

The correct answer is: `setup_local_APIC()`

Question **26**
Correct
Mark 5.00 out of 5.00

Intel keyboard controller is found on the motherboard. The controller handles input received from the computer keyboard, A20 lines, reset, deciphering scan codes, as well as the PS/2 mouse.

Select one:

- ☐ 1. 8259
- ☐ 2. 8048
- ☐ 3. 8051
- ☒ 4. 8042 ✓

The correct answer is: **8042**

Question **27**
Correct
Mark 5.00 out of 5.00

..... is used to Transmit packets to network cards

Select one:

- ☐ 1. `NET_RX_SOFTIRQ`
- ☐ 2. `SCSI_SOFTIRQ`
- ☐ 3. `TASKLET_SOFTIRQ`
- ☒ 4. `NET_TX_SOFTIRQ` ✓

The correct answer is: `NET_TX_SOFTIRQ`

Question **28**

Correct

Mark 5.00 out of 5.00

Say TRUE or FALSE -

" Each core in the processor has a separate APR & TPR Registers"

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **29**

Correct

Mark 5.00 out of 5.00

Keyboard Controller Data Register port address is

Select one:

- ☒ 1. 0x60 ✓
- ☐ 2. 0x84
- ☐ 3. 0x80
- ☐ 4. 0x64

The correct answer is: 0x60

Question **30**

Correct

Mark 5.00 out of 5.00

The interrupt request on INTR pin can be disabled by resetting the following flag in the processor status register:

Select one:

- ☐ 1. DF
- ☐ 2. OF
- ☐ 3. ZF
- ☒ 4. IF ✓

The correct answer is: IF

Question **31**

Correct

Mark 5.00 out of 5.00

If an exception is generated by the execution of an instruction then it is called interrupt.

Select one:

- ☐ 1. asynchronous
- ☒ 2. synchronous ✓
- ☐ 3. unacceptable
- ☐ 4. unhandleable

The correct answer is: synchronous

Question **32**

Incorrect

Mark 0.00 out of 5.00

Identify whether the exception **Overflow** is

Select one:

- ☐ 1. abort
- ☐ 2. interrupt
- ☒ 3. fault 
- ☐ 4. trap

The correct answer is: trap

Question **33**

Correct

Mark 5.00 out of 5.00

..... Receives packets from network cards

Select one:

- ☐ 1. TASKLET_SOFTIRQ
- ☒ 2. NET_RX_SOFTIRQ 
- ☐ 3. NET_TX_SOFTIRQ
- ☐ 4. SCSI_SOFTIRQ

The correct answer is: NET_RX_SOFTIRQ

Question **34**

Correct

Mark 5.00 out of 5.00

An interrupt breaks the execution of instructions and diverts its execution to

Select one:

- ☐ 1. Control Unit
- ☐ 2. Counter word register
- ☒ 3. Interrupt service routine 
- ☐ 4. Arithmetic Logic Unit

The correct answer is: Interrupt service routine

Question **35**

Correct

Mark 5.00 out of 5.00

..... command will list modules that are already loaded in the kernel.

Answer:



The correct answer is: lsmod

Question **36**

Correct

Mark 5.00 out of 5.00

Identify whether the exception Divide error is

Select one:

- ☐ 1. interrupt
- ☐ 2. trap
- ☒ 3. fault ✓
- ☐ 4. abort

The correct answer is: fault

Question **37**

Correct

Mark 5.00 out of 5.00

Assume that in Linux system, the processor raised "Double Fault" Exception when executing an instruction.

What is the vector associated with the Double Fault exception?

Answer:



The correct answer is: 8

Question **38**

Correct

Mark 5.00 out of 5.00

The int vector associated with the NMI is

Select one:

- ☒ 1. 2 ✓
- ☐ 2. 8
- ☐ 3. 4
- ☐ 4. 14

The correct answer is: 2

Question **39**

Correct

Mark 5.00 out of 5.00

Example of an synchronous interrupt is

Select one:

- ☐ 1. keyboard interrupt
- ☐ 2. NMI
- ☒ 3. divide by zero interrupt ✓
- ☐ 4. INTR

The correct answer is: divide by zero interrupt

Question **40**

Correct

Mark 5.00 out of 5.00

IRQ number associated with keyboard is

Select one:

- ☒ 1. 1 ✓
- ☐ 2. 32
- ☐ 3. 33
- ☐ 4. 0

The correct answer is: 1

Question **41**

Incorrect

Mark 0.00 out of 5.00

Assume that in Linux system, the processor raised "Double Fault" Exception when executing an instruction.

What is the type of the gate descriptor stored in that entry in the IDT?

Answer: ✗

The correct answer is: Task gate

Question **42**

Correct

Mark 10.00 out of 10.00

When the device operation is concluded, the driver invokes the function to remove the descriptor from the IRQ list and release the memory area.

Answer: ✓

The correct answer is: free_irq()

Question **43**

Correct

Mark 5.00 out of 5.00

The int vector associated with the exception "Page fault" is

Select one:

- ☐ 1. 5
- ☐ 2. 8
- ☐ 3. 4
- ☒ 4. 14 ✓

The correct answer is: 14

Question **44**

Correct

Mark 5.00 out of 5.00

On multiprocessor systems, vector is sent to all CPUs but the sender, forcing those CPUs to run a function passed by the sender. The corresponding interrupt handler is named `call_function_interrupt()`.

Select one:

- ☐ 1. INVALIDATE_TLB_VECTOR (vector 0xfd)
- ☐ 2. RESCHEDULE_VECTOR (vector 0xfc)
- ☐ 3. int 128
- ☒ 4. CALL_FUNCTION_VECTOR (vector 0xfb) ✓

The correct answer is: CALL_FUNCTION_VECTOR (vector 0xfb)

Question **45**

Incorrect

Mark 0.00 out of 5.00

In Linux, breakpoint exception handler is activated using which of the following :

Select one:

- ☐ 1. System gate
- ☒ 2. Trap gate ✗
- ☐ 3. Interrupt gate
- ☐ 4. System Interrupt gate

The correct answer is: System Interrupt gate

Question **46**

Correct

Mark 5.00 out of 5.00

..... are piece of code, that can be loaded and unloaded from kernel on demand.

Answer: ✓

The correct answer is: Kernel modules

Question **47**

Correct

Mark 10.00 out of 10.00

To allow arguments to be passed to Kernel module, declare the variables that will take the values of the command line arguments as global and then use the macro, to set the mechanism up.

Answer: ✓

The correct answer is: module_param()

Question **48**

Correct

Mark 5.00 out of 5.00

Select the interrupt class to which the following actions belong :

"Actions such as acknowledging an interrupt to the PIC, reprogramming the PIC or the device controller, or updating data structures accessed by both the device and the processor."

Select one:

- ☐ 1. NonCritical Deferrable
- ☐ 2. NonCritical
- ☒ 3. Critical ✓
- ☐ 4. I/O Interrupts

The correct answer is: Critical

Question **49**

Partially correct

Mark 10.00 out of 15.00

Linux divides the actions to be performed following an interrupt into three classes. Identify them in the following:

Select one or more:

- ☒ 1. Noncritical Deferrable ✓
- ☐ 2. Timer Interrupts
- ☐ 3. InterProcessor Interrupts
- ☒ 4. Critical ✓
- ☒ 5. I/O Interrupts ✗
- ☐ 6. Noncritical

The correct answers are: Critical, Noncritical, Noncritical Deferrable

Question **50**

Correct

Mark 5.00 out of 5.00

Interrupt vectors normally associated with physical IRQs -

Select one:

- ☐ 1. 20-31
- ☐ 2. 0-19
- ☐ 3. 251-253
- ☒ 4. 32-127 ✓

The correct answer is: 32-127

Question **51**

Correct

Mark 5.00 out of 5.00

..... allow deferrable function in the context of process to be activated and later executed by special kernel threads

Select one:

- ☐ 1. Traps
- ☐ 2. Tasklets
- ☒ 3. Work queues ✓
- ☐ 4. Softirqs

The correct answer is: Work queues

Question **52**

Correct

Mark 5.00 out of 5.00

..... softirq Handles high priority tasklets

Select one:

- ☐ 1. TASKLET_SOFTIRQ
- ☒ 2. HI_SOFTIRQ ✓
- ☐ 3. NET_TX_SOFTIRQ
- ☐ 4. Timer_SOFTIRQ

The correct answer is: HI_SOFTIRQ

Question **53**

Correct

Mark 5.00 out of 5.00

.....actions are executed within the interrupt handler immediately, with maskable interrupts disabled.

Answer:



The correct answer is: Critical

Question **54**

Incorrect

Mark 0.00 out of 5.00

Interrupt vectors normally associated with Non-maskable interrupts and exceptions -

Select one:

- ☐ 1. 32-127
- ☐ 2. 0-19
- ☐ 3. 20-31
- ☒ 4. 251-253 ✗

The correct answer is: 0-19

Question **55**
Correct
Mark 10.00 out of 10.00

When a module is inserted into the kernel, the macro will be invoked, which will call the __init function

Answer: ✓

The correct answer is: module_init

Question **56**
Incorrect
Mark 0.00 out of 5.00

..... is a function that prints messages and is used in the C Programming Language exclusively for the Linux Kernel.

Answer: ✗

The correct answer is: printk

Question **57**
Correct
Mark 5.00 out of 5.00

Number of Interrupt Request IRQ lines on I/O APIC is

Select one:

- ☐ 1. 8
- ☐ 2. 32
- ☒ 3. 24 ✓
- ☐ 4. 16

The correct answer is: 24

Question **58**
Correct
Mark 5.00 out of 5.00

Say TRUE or FALSE -

" Each core in the processor has a separate local APIC".

Select one:

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question **59**

Correct

Mark 5.00 out of 5.00

The int vector associated with the System Call is

Select one:

- ☒ 1. 128 ✓
- ☐ 2. 14
- ☐ 3. 4
- ☐ 4. 8

The correct answer is: 128

Question **60**

Incorrect

Mark 0.00 out of 5.00

..... is an intelligent command which will load/unload modules based on the dependency between modules.

Answer:



The correct answer is: modprobe

Question **61**

Not answered

Marked out of 15.00

Interrupt handling depends on the type of the interrupt. There are three main classes of Interrupts - Identify them in the following:

Select one or more:

- ☐ 1. Interprocessor Interrupts
- ☐ 2. Non-Critical
- ☐ 3. Critical
- ☐ 4. Timer Interrupts
- ☐ 5. I/O Interrupts
- ☐ 6. Non Critical Deferrable

The correct answers are: I/O Interrupts, Timer Interrupts, Interprocessor Interrupts

Question **62**

Not answered

Marked out of 5.00

Intel microcontroller inside the keyboard is used to scan the keyboard to detect the key being pressed or released and returns the scan code of the key being pressed or released.

Select one:

- ☐ 1. 8042
- ☐ 2. 8259
- ☐ 3. 8048
- ☐ 4. 8251

The correct answer is: 8048

Question **63**

Correct

Mark 5.00 out of 5.00

Identify whether the exception Machine Check is

Select one:

- ☐ 1. trap
- ☐ 2. interrupt
- ☒ 3. abort ✓
- ☐ 4. fault

The correct answer is: abort

Question **64**

Correct

Mark 10.00 out of 10.00

IRQ number and int vector associated with the hardware device Keyboard

Select one:

- ☐ 1. 13, 45
- ☒ 2. 1, 33 ✓
- ☐ 3. 10, 42
- ☐ 4. 0, 32

The correct answer is: 1, 33

Question **65**

Correct

Mark 5.00 out of 5.00

..... stack is used when handling Non-critical deferrable functions

Select one:

- ☒ 1. Soft IRQ ✓
- ☐ 2. System
- ☐ 3. Hard IRQ
- ☐ 4. Exception

The correct answer is: Soft IRQ

Question **66**

Correct

Mark 10.00 out of 10.00

..... function creates a new irqaction descriptor and initializes it with the parameter values; it then invokes the setup_x86_irq() function to insert the descriptor in the proper IRQ list.

Answer:



The correct answer is: request_irq

Question **67**

Correct

Mark 5.00 out of 5.00

The Programmable interrupt controller is required to

Select one:

- ☐ 1. handle one interrupt request
- ☐ 2. handle no interrupt request
- ☒ 3. handle one or more interrupt requests at a time ✓
- ☐ 4. handle 32 interrupt requests with a delay

The correct answer is: handle one or more interrupt requests at a time

Question **68**

Correct

Mark 10.00 out of 10.00

Number of entries and size of each entry in IDT are & respectively

Select one:

- ☐ 1. 64 & 4 bytes
- ☒ 2. 256 & 8 bytes ✓
- ☐ 3. 256 & 64 bytes
- ☐ 4. 128 & 8 bytes

The correct answer is: 256 & 8 bytes

Question **69**

Correct

Mark 5.00 out of 5.00

..... actions are executed by the interrupt handler immediately, with the interrupts enabled.

Answer: ✓

The correct answer is: Non-Critical

Question **70**

Not answered

Marked out of 5.00

Select the interrupt class to which the following actions belong :

"Actions such as updating data structures that are accessed only by the processor (for instance, reading the scan code after a keyboard key has been pushed)."

Select one:

- ☐ 1. Timer Interrupts
- ☐ 2. NonCritical Deferrable
- ☐ 3. Critical
- ☐ 4. NonCritical

The correct answer is: NonCritical

Question **71**
Not answered
Marked out of 10.00

Two main functions required in any Kernel Module are

Select one:

- ☐ 1. __init, --exit
- ☐ 2. --init, --exit
- ☐ 3. --init, __exit
- ☐ 4. __init, __exit

The correct answer is: __init, __exit

Question **72**
Correct
Mark 5.00 out of 5.00

Inexception, the saved value of eip is the address of the instruction that caused the exception, and hence that instruction can be resumed when the exception handler terminates.

Select one:

- ☐ 1. Trap
- ☒ 2. Fault ✓
- ☐ 3. Abort
- ☐ 4. Machine Check

The correct answer is: Fault

Question **73**
Not answered
Marked out of 5.00

..... command will display information about a kernel module

Answer:



The correct answer is: modinfo

Question **74**
Not answered
Marked out of 5.00

The kernel modules will have a extension.

Select one:

- ☐ 1. .ko
- ☐ 2. .c
- ☐ 3. .cpp
- ☐ 4. .exe

The correct answer is: .ko

Question **75**

Correct

Mark 5.00 out of 5.00

In exception, the saved value of eip is the address of the instruction that should be executed after the one that caused the exception.

Select one:

- ☐ 1. Abort
- ☐ 2. Machine Check
- ☒ 3. Trap ✓
- ☐ 4. Fault

The correct answer is: Trap

Question **76**

Correct

Mark 5.00 out of 5.00

The device permits its IRQ line to be shared with other devices when the flag

Select one:

- ☐ 1. IRQF_INTERRUPT=1
- ☒ 2. IRQF_SHIRQ=1 ✓
- ☐ 3. IRQF_SHIRQ=0
- ☐ 4. IRQF_INTERRUPT=0

The correct answer is: IRQF_SHIRQ=1

Question **77**

Correct

Mark 10.00 out of 10.00

The kernel thread kirqd exploits a nice feature of multi-APIC systems, called the of a CPU, by modifying the Interrupt Redirection Table entries of the I/O APIC, it is possible to route an interrupt signal to a specific CPU.

Answer: ✓

The correct answer is: IRQ affinity

Question **78**

Correct

Mark 5.00 out of 5.00

..... register of CPU contains the initial address of IDT table

Select one:

- ☐ 1. gdtr
- ☐ 2. cs
- ☐ 3. ldtr
- ☒ 4. idtr ✓

The correct answer is: idtr

Question **79**

Not answered
Marked out of
5.00

..... field of the irqaction descriptor points to the descriptor of the /proc/irq/n directory associated with the IRQn

Answer:



The correct answer is: dir

Question **80**

Not answered
Marked out of
5.00

..... specifies the type of message being sent to the kernel message log.

Answer:



The correct answer is: Loglevel

Question **81**

Not answered
Marked out of
10.00

..... generates an updated dependency list to make sure that the module we're installing is aware of all existing modules and dependencies. This dependency list will be used by modprobe when installing the kernel module

Answer:



The correct answer is: sudo depmod -a

Question **82**

Correct
Mark 10.00 out
of 10.00

If the segment selector in the gate descriptor contains the value 0x78, then determine which entry of the GDT is accessed to get the base address of the corresponding linear segment containing the ISR.

Answer: 0xF



The correct answer is: 0xF

Question **83**

Correct
Mark 5.00 out
of 5.00

Identify whether the exception Double Fault is

Select one:

- ☐ 1. fault
- ☐ 2. trap
- ☐ 3. interrupt
- ☒ 4. abort ✓

The correct answer is: abort

Question **84**

Incorrect

Mark 0.00 out of 5.00

Identify whether the exception Bounds check is

Select one:

- ☐ 1. trap
- ☐ 2. interrupt
- ☒ 3. abort 
- ☐ 4. fault

The correct answer is: fault

Question **85**

Not answered

Marked out of 5.00

..... command will remove a module from the kernel. You cannot remove a module which is already used by any program.

Answer: 

The correct answer is: rmmod

Question **86**

Correct

Mark 5.00 out of 5.00

An interrupt is if it is not handled by the kernel, that is, either if there is no ISR associated with the IRQ line, or if no ISR associated with the line recognizes the interrupt as raised by its own hardware device.

Answer: 

The correct answer is: unexpected

Question **87**

Correct

Mark 5.00 out of 5.00

..... actions are executed within the interrupt handler immediately, with maskable interrupts disabled

Answer: 

The correct answer is: Critical

Question **88**

Not answered

Marked out of 5.00

Most exceptions are handled simply by sending a Unixto the process that caused the exception.

Answer: 

The correct answer is: signal

Question **89**

Correct

Mark 10.00 out of 10.00

The kernel stores in the `irq_count` and `irqs_unhandled` fields of the `irq_desc_t` descriptor the total number of interrupts and the unexpected interrupts, respectively, when the interrupt is raised, the kernel disables the line if the number of unhandled interrupts is above

Select one:

- ☐ 1. 1000,000th, 999,900
- ☒ 2. 100,000th, 99,900 ✓
- ☐ 3. 10,000th, 9,900
- ☐ 4. 1,000th, 900

The correct answer is: 100,000th, 99,900

Question **90**

Correct

Mark 10.00 out of 10.00

Noncritical deferrable actions are performed by means of separate functions called as...

Select one:

- ☐ 1. Intirqs & Tasklets
- ☒ 2. Softirqs and Tasklets ✓
- ☐ 3. Softlets and Taskirqs
- ☐ 4. Hardirqs & Tasklets

The correct answer is: Softirqs and Tasklets

Question **91**

Correct

Mark 5.00 out of 5.00

Intel's default vector associated with IRQn line is

Select one:

- ☐ 1. $n + 8$
- ☐ 2. $n + 64$
- ☒ 3. $n + 32$ ✓
- ☐ 4. $n + 16$

The correct answer is: $n + 32$

Question **92**

Correct

Mark 5.00 out of 5.00

The int vector associated with the exception "Page fault" is

Select one:

- ☐ 1. 8
- ☒ 2. 14 ✓
- ☐ 3. 4
- ☐ 4. 3

The correct answer is: 14

Question **93**
Not answered
Marked out of 5.00

..... Vector range is used for Non-maskable Interrupt and exceptions

Select one:

- ☐ 1. 128 (0x80)
- ☐ 2. 0 - 19(0x0 - 0x13)
- ☐ 3. 251 - 253(0xfb - 0xfd)
- ☐ 4. 32 - 127(0x20 - 0x7f)

The correct answer is: 0 - 19(0x0 - 0x13)

Question **94**
Not answered
Marked out of 5.00

Indicate the total number of Loglevels which can be specified in printf functions.

Select one:

- ☐ 1. 32
- ☐ 2. 4
- ☐ 3. 16
- ☐ 4. 8

The correct answer is: 8

Question **95**
Correct
Mark 5.00 out of 5.00

Which entry of the GDT is accessed to get the base address of the linear segment containing the Exception Handler associated with the Double Fault exception?

Answer: ✓

The correct answer is: 31

Question **96**
Correct
Mark 10.00 out of 10.00

APR stands for

Answer: ✓

The correct answer is: Arbitration Priority Register

Question **97**
Correct
Mark 10.00 out of 10.00

IPI stands for

Answer: ✓

The correct answer is: Inter Processor Interrupt

Question **98**

Correct

Mark 5.00 out of 5.00

An interrupt is said to be if it is not handled by the kernel, that is, either if there is no ISR associated with the IRQ line, or if no ISR associated with the line recognizes the interrupt as raised by its own hardware device.

Select one:

- ☐ 1. external
- ☐ 2. asynchronous
- ☒ 3. unexpected ✓
- ☐ 4. synchronous

The correct answer is: unexpected

Question **99**

Correct

Mark 5.00 out of 5.00

Which flag of the processor is used to disable or enable maskable Interrupts?

Select one:

- ☐ 1. DF
- ☐ 2. OF
- ☐ 3. ZF
- ☒ 4. IF ✓

The correct answer is: IF

Question **100**

Correct

Mark 5.00 out of 5.00

NMI stands for

Select one:

- ☐ 1. New Modern Interrrupt
- ☒ 2. Non Maskable Interrupt ✓
- ☐ 3. Non Movable interrupt
- ☐ 4. Non Multiple Interrupt

The correct answer is: Non Maskable Interrupt

Question **101**

Correct

Mark 10.00 out of 10.00

I/O APIC stands for

Select one:

- ☐ 1. Input/Output Advanced Processor Interrupt Controller
- ☐ 2. Input/Output Advanced Processor Interleaved Cache
- ☐ 3. Input/Output Advanced Peripheral Interface Circuit
- ☒ 4. Input/Output Advanced Programmable Interrupt Controller ✓

The correct answer is: Input/Output Advanced Programmable Interrupt Controller

Question **102**

Correct

Mark 5.00 out of 5.00

..... Softirq handles tasklets related to timer interrupts

Select one:

- ☒ 1. Timer_SOFTIRQ ✓
- ☐ 2. NET_TX_SOFTIRQ
- ☐ 3. SCSI_SOFTIRQ
- ☐ 4. NET_RX_SOFTIRQ

The correct answer is: Timer_SOFTIRQ

Question **103**

Correct

Mark 5.00 out of 5.00

The kernel thread periodically executes the `do_irq_balance()` function, which keeps track of the number of interrupt occurrences received by every CPU in the most recent time interval.

Answer:

The correct answer is: kirqd

Question **104**

Correct

Mark 5.00 out of 5.00

Kernel makes use of Soft IRQ stack for each CPU in the system and each stack is contained in a single page frame.

Select one:

- ☐ 1. 4
- ☐ 2. 2
- ☒ 3. 1 ✓
- ☐ 4. 3

The correct answer is: 1

Question **105**

Correct

Mark 5.00 out of 5.00

The int vector associated with the exception "Breakpoint " is

Select one:

- ☐ 1. 14
- ☐ 2. 4
- ☐ 3. 8
- ☒ 4. 3 ✓

The correct answer is: 3

Question **106**

Correct

Mark 5.00 out of 5.00

The interrupt for which the processor has highest priority among all the external interrupts is

Select one:

- ☐ 1. TRAP
- ☐ 2. INTR
- ☐ 3. keyboard interrupt
- ☒ 4. NMI ✓

The correct answer is: NMI

Question **107**

Not answered

Marked out of 5.00

..... vector range is used for External Interrupts (IRQs)

Select one:

- ☐ 1. 0 - 19(0x0 - 0x13)
- ☐ 2. 20 - 31(0x14 - 0x1f)
- ☐ 3. 251 - 253(0xfb - 0xfd)
- ☐ 4. 32 - 127(0x20 - 0x7f) & 129 - 238(0x81 - 0xee)

The correct answer is: 32 - 127(0x20 - 0x7f) & 129 - 238(0x81 - 0xee)

Question **108**

Correct

Mark 10.00 out of 10.00

Interrupt vectors normally associated with Inter-processor interrupts -

Select one:

- ☒ 1. 251-253 ✓
- ☐ 2. 32-127
- ☐ 3. 20-31
- ☐ 4. 0-19

The correct answer is: 251-253

Question **109**

Correct

Mark 5.00 out of 5.00

..... can run concurrently on several CPUs, even if they are of the same type.

Answer: Softriqs

The correct answer is: Softriqs

Question **110**

Correct

Mark 10.00 out of 10.00

Interrupt arriving on.....pin on the processor can be disabled by clearing the.....bit in the eflags register

Select one:

- ☐ 1. NMI, IF
- ☐ 2. NMI, TF
- ☒ 3. INTR, IF ✓
- ☐ 4. INTR, TF

The correct answer is: INTR, IF

Question **111**

Correct

Mark 5.00 out of 5.00

The consists of a set of 24 IRQ lines, a 24 entry Interrupt Redirection table, programmable registers, and a message unit for sending and receiving APIC messages over the APIC bus.

Answer: ✓

The correct answer is: I/O APIC

Question **112**

Correct

Mark 5.00 out of 5.00

How many interrupting devices one 8259 PIC chip can support?

Select one:

- ☐ 1. 24
- ☐ 2. 16
- ☒ 3. 8 ✓
- ☐ 4. 4

The correct answer is: 8

Question **113**

Correct

Mark 5.00 out of 5.00

The int vector associated with the exception "Breakpoint " is

Select one:

- ☐ 1. 4
- ☐ 2. 8
- ☒ 3. 3 ✓
- ☐ 4. 14

The correct answer is: 3

Question **114**
Not answered
Marked out of 5.00

Compute the total memory space required to store IDT in memory.....

Select one:

- ☐ 1. 4MB
- ☐ 2. 2KB
- ☐ 3. 4KB
- ☐ 4. 2MB

The correct answer is: 2KB

Question **115**
Not answered
Marked out of 5.00

..... command will insert a new module into the kernel

Answer:



The correct answer is: insmod

Question **116**
Not answered
Marked out of 5.00

..... vector range is used for Interprocessor Interrupts

Select one:

- ☐ 1. 129 - 238(0x81 - 0xee)
- ☐ 2. 241 - 250(0xf1 - 0xfa)
- ☐ 3. 0 - 19(0x0 - 0x13)
- ☐ 4. 251 - 253(0xfb - 0xfd)

The correct answer is: 251 - 253(0xfb - 0xfd)

Question **117**
Not answered
Marked out of 10.00

..... Loglevel in printk indicates

"Some problem has occurred, immediate attention is needed"

Select one:

- ☐ 1. KERN_EMERG
- ☐ 2. KERN_ALERT
- ☐ 3. KERN_CRIT
- ☐ 4. KERN_ERR

The correct answer is: KERN_ALERT

Question **118**

Correct

Mark 10.00 out of 10.00

All exception handlers(except breakpoint, overflow, bound check & syscall) are activated bydescriptor and executed in mode.

Select one:

- ☐ 1. Trap gate, user
- ☒ 2. Trap gate, kernel ✓
- ☐ 3. System Interrupt gate, kernel
- ☐ 4. System gate, user

The correct answer is: Trap gate, kernel

Question **119**

Not answered

Marked out of 5.00

The device driver aborts the operation if returns an error code, which means that the IRQ line is already in use by another device that does not allow interrupt sharing.

Select one:

- ☐ 1. setup_x86_irq()
- ☐ 2. setup_local_APIC()
- ☐ 3. do_irq_balance()
- ☐ 4. set_ioapic_affinity_irq()

The correct answer is: setup_x86_irq()

Question **120**

Correct

Mark 10.00 out of 10.00

TPR stands for

Answer: ✓

The correct answer is: Task Priority Register

Question **121**

Correct

Mark 10.00 out of 10.00

ICC stands for

Answer: ✓

The correct answer is: Interrupt Controller Communication

Question **122**

Correct

Mark 5.00 out of 5.00

..... stack is used when handling interrupts (critical and Non-critical interrupt functions)

Select one:

- ☐ 1. System
- ☐ 2. Soft IRQ
- ☐ 3. Exception
- ☒ 4. Hard IRQ ✓

The correct answer is: Hard IRQ

Question **123**

Correct

Mark 5.00 out of 5.00

....., stack is used when handling exceptions (including system calls)

Select one:

- ☐ 1. Hard IRQ
- ☒ 2. Exception ✓
- ☐ 3. System
- ☐ 4. Soft IRQ

The correct answer is: Exception

Question **124**

Correct

Mark 5.00 out of 5.00

In Linux , syscall exception handlers are activated using which of the following :

Select one:

- ☐ 1. Interrupt gate
- ☐ 2. Trap gate
- ☐ 3. System Interrupt gate
- ☒ 4. System gate ✓

The correct answer is: System gate

Question **125**

Correct

Mark 5.00 out of 5.00

The int vector associated with the exception " Bounds check" is

Select one:

- ☐ 1. 8
- ☐ 2. 14
- ☒ 3. 5 ✓
- ☐ 4. 4

The correct answer is: 5

Question **126**

Correct

Mark 5.00 out of 5.00

Identify whether the exception Breakpoint is

Select one:

- ☐ 1. interrupt
- ☐ 2. fault
- ☒ 3. trap ✓
- ☐ 4. abort

The correct answer is: trap

Question **127**

Correct

Mark 5.00 out of 5.00

Example of a synchronous interrupt is

Select one:

- ☐ 1. type2 interrupt
- ☒ 2. double fault interrupt ✓
- ☐ 3. keyboard interrupt
- ☐ 4. timer interrupt

The correct answer is: double fault interrupt

Question **128**

Correct

Mark 5.00 out of 5.00

If an interrupt is generated from outside the CPU then it is an

Select one:

- ☒ 1. asynchronous ✓
- ☐ 2. synchronous
- ☐ 3. unhandleable
- ☐ 4. unexpected

The correct answer is: asynchronous

Question **129**

Not answered

Marked out of 10.00

The C function used to read a byte from either data register or status register of a keyboard controller by specifying its corresponding port address is

Select one:

- ☐ 1. outputb(port_address)
- ☐ 2. inb(port_address)
- ☐ 3. inputb(port_address)
- ☐ 4. outb(port_address)

The correct answer is: inb(port_address)

Question **130**

Correct

Mark 10.00 out of 10.00

Given that idtr register contains the address 0x00010000 and the vector received from the Interrupting device is 37. Determine the address of the gate descriptor (in hex) in the IDT corresponding to the interrupt vector 37.

Answer:



The correct answer is: 0x00010128

Question **131**

Correct

Mark 5.00 out of 5.00

Each entry in the in I/O APIC can be individually programmed to indicate vector and priority, the destination processor, and how the processor is selected.

Answer:



The correct answer is: Interrupt Redirection Table

Question **132**

Correct

Mark 5.00 out of 5.00

The interrupt for which the processor has highest priority among all the internal interrupts is

Select one:

- ☐ 1. keyboard interrupt
- ☒ 2. TRAP
- ☐ 3. NMI
- ☐ 4. INT

The correct answer is: TRAP

Question **133**

Not answered

Marked out of 10.00

All kernel messages, that is, outputs from printk in kernel module are routed to a kernel log file(specify the file name with complete pathname)

Answer:



The correct answer is: /var/log/kern.log

Question **134**

Correct

Mark 10.00 out of 10.00

IRQ number and int vector are associated with the hardware device Timer

Select one:

- ☐ 1. 1, 33
- ☐ 2. 10, 42
- ☐ 3. 13, 45
- ☒ 4. 0, 32 ✓

The correct answer is: 0, 32

Question **135**

Not answered

Marked out of 5.00

The kernel maintains descriptors, each of which refers to a specific hardware device and a specific interrupt.

Answer:



The correct answer is: irqaction

Question **136**

Not answered

Marked out of 5.00

Select the interrupt class to which the following actions belong :

"Actions such as copying a buffer's contents into the address space of some process (for instance, sending the keyboard line buffer to the terminal handler process)."

Select one:

- ☐ 1. Timer Interrupts
- ☐ 2. Critical
- ☐ 3. NonCritical Deferrable
- ☐ 4. NonCritical

The correct answer is: NonCritical

Question **137**

Not answered

Marked out of 5.00

The int vector associated with the exception "Overflow" is

Select one:

- ☐ 1. 5
- ☐ 2. 8
- ☐ 3. 4
- ☐ 4. 14

The correct answer is: 4

Question **138**

Correct

Mark 5.00 out of 5.00

Say TRUE or FALSE -

"Interrupt request on NMI pin on the processor can be disabled by resetting the IF flag of the processor status register".

Select one:

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question **139**

Correct

Mark 5.00 out of 5.00

In Linux , external interrupt handlers are activated using which of the following :

Select one:

- ☐ 1. System Interrupt gate
- ☐ 2. Trap gate
- ☒ 3. Interrupt gate ✓
- ☐ 4. System gate

The correct answer is: Interrupt gate

Question **140**

Correct

Mark 5.00 out of 5.00

Example of an asynchronous interrupt is

Select one:

- ☐ 1. divide by zero interrupt
- ☒ 2. keyboard interrupt ✓
- ☐ 3. breakpoint interrupt
- ☐ 4. overflow interrupt

The correct answer is: keyboard interrupt

Question **141**

Incorrect

Mark 0.00 out of 10.00

Assume that in Linux system, the processor raised "Double Fault" Exception when executing an instruction.

What is the value (in hex) stored in the Segment Selector in that gate entry of the IDT?

Answer: ✗

The correct answer is: 0xF8

Question **142**

Correct

Mark 10.00 out of 10.00

Given that idtr register contains the address 0x00020000 and the vector received from the Interrupting device is 40. Determine the address of the gate descriptor in the IDT corresponding to the interrupt vector 40.

Select one:

- ☐ 1. 0x00020200
- ☒ 2. 0x00020140 ✓
- ☐ 3. 0x00020040
- ☐ 4. 0x00020320

The correct answer is: 0x00020140

Question **143**

Correct

Mark 5.00 out of 5.00

If the hardware fails to distribute the interrupts among the CPUs in a fair way, Linux makes use of a special kernel thread calledto correct, if necessary, the automatic assignment of IRQs to CPUs.

Select one:

- ☒ 1. kirqd ✓
- ☐ 2. nfsd
- ☐ 3. crond
- ☐ 4. syncd

The correct answer is: kirqd

Question **144**

Correct

Mark 5.00 out of 5.00

Softirqs are functions and must explicitly protect their data structures with spin locks.

Answer: ✓

The correct answer is: reentrant

Question **145**

Not answered

Marked out of 10.00

Determine the number of Intel 8259 PICs required to handle a maximum of 64 Interrupting I/O Devices.

Answer: ✗

The correct answer is: 9